Comparing Numerical Solvers and Physics-Informed Neural Networks for the Damped Harmonic Oscillator

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Introduction

The damped harmonic oscillator is a classical dynamical system that appears in many physical and engineering applications, including mechanical vibration, structural dynamics, and electrical circuits. Its behavior is governed by the second-order ordinary differential equation

$$m\ddot{x} + c\dot{x} + kx = 0,$$

where m is the mass, c is the damping coefficient, and k is the spring constant. Because the damping term dissipates energy, the oscillation amplitude decreases over time.

This project compares three approaches for solving the damped harmonic oscillator: the explicit Euler method, the fourth-order Runge–Kutta (RK4) method, and a simple Physics-Informed Neural Network (PINN). The goal is to evaluate their accuracy and discuss the advantages and limitations of classical numerical solvers versus scientific machine learning methods.

Method

In this project, the parameters were chosen as

$$m = 1.0, \quad c = 0.4, \quad k = 4.0,$$

with initial conditions

$$x(0) = 1.0, \quad \dot{x}(0) = 0.0.$$

The time interval was set to $t \in [0, 10]$.

To solve the problem numerically, the second-order equation was rewritten as a first-order system:

$$x' = v,$$

$$v' = -\frac{c}{m}v - \frac{k}{m}x.$$

A reference solution was first generated using `solve_ivp` from SciPy. This reference was used to evaluate the accuracy of the other methods.

Two classical numerical solvers were implemented: the explicit Euler method and RK4. Their predicted displacement curves were compared against the reference solution, and their mean squared errors (MSE) were computed.

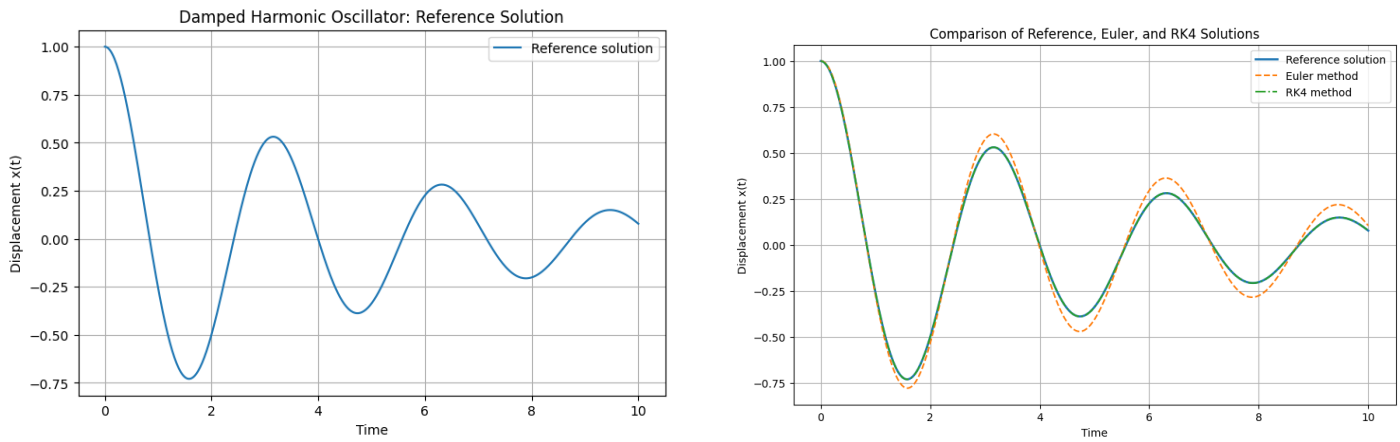
A simple PINN was then constructed using a fully connected neural network with two hidden layers and `tanh` activation functions. Instead of relying on labeled training data, the PINN was trained by minimizing a loss function that included both the differential equation residual and the initial condition constraints:

$$\mathcal{L} = \mathcal{L}_{\text{physics}} + \mathcal{L}_{\text{IC}}.$$

This allowed the network to learn a function $x(t)$ that approximately satisfies the governing equation and initial conditions.

Results

Figure 1 summarizes the baseline and classical numerical results. Figure 1a shows the reference solution of the damped harmonic oscillator, which exhibits underdamped oscillatory behavior with gradually decaying amplitude. Figure 1b compares the Euler and RK4 solutions with the reference solution. Both methods capture the overall oscillatory trend, but RK4 tracks the reference much more closely than Euler.



(a) Reference solution.

(b) Reference, Euler, and RK4 comparison.

Figure 1: Reference solution and comparison of classical numerical solvers.

The numerical MSE values were:

$$\text{Euler MSE} \approx 2.51 \times 10^{-3},$$

$$\text{RK4 MSE} \approx 3.04 \times 10^{-7}.$$

These results confirm that RK4 is significantly more accurate than Euler for this problem.

For the PINN, the training loss decreased steadily during optimization, indicating that the model was learning to satisfy both the governing equation and the initial conditions. However, the final PINN prediction deviated noticeably from the reference solution, especially at later times. The PINN MSE was

$$\text{PINN MSE} \approx 2.06 \times 10^{-2},$$

which was larger than both Euler and RK4 in this experiment.

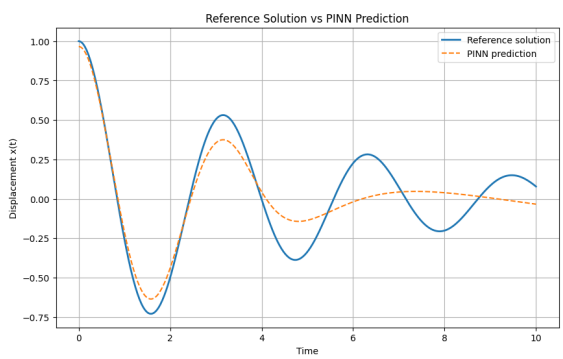


Figure 2: Comparison of the reference solution and PINN prediction.

Discussion

Among the three tested methods, RK4 produced the most accurate solution by a large margin. Euler approximated the overall dynamics but accumulated larger errors over time. The basic PINN captured the qualitative damped oscillatory trend, but its quantitative accuracy was limited.

This is reasonable because RK4 is a high-order solver designed for ODEs, whereas the PINN learns the solution through optimization. For a simple low-dimensional ODE such as the damped harmonic oscillator, classical solvers remain more efficient and reliable. However, PINNs are still important because they incorporate physical laws into learning and may be more useful when data are sparse, equations are partially known, or traditional solvers are difficult to apply.

Conclusion

This project compared Euler, RK4, and a simple PINN for solving the damped harmonic oscillator. RK4 achieved the highest accuracy, Euler showed moderate accuracy, and the PINN captured the qualitative behavior but had the largest error. These results show that classical numerical methods remain highly effective for simple ODEs, while PINNs provide a flexible machine-learning framework that may become more competitive with further tuning or in more complex settings.